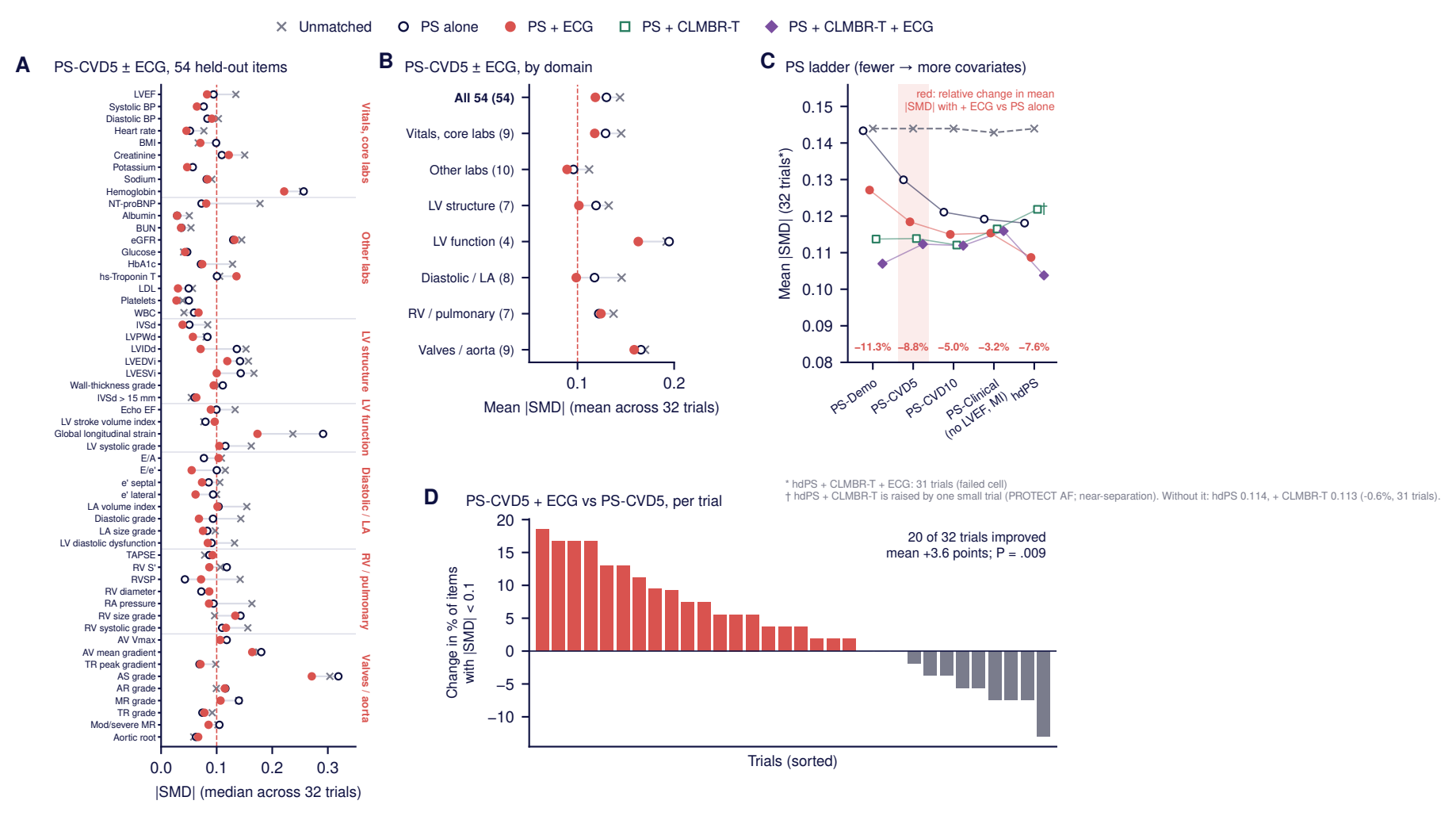
Unmatched
PS alone
PS + ECG
PS + CLMBR-T
PS + CLMBR-T + ECG
A
PS-CVD5 ± ECG, 54 held-out items
LVEF
Systolic BP
Diastolic BP
Heart rate
BMI
Creatinine
Potassium
Sodium
Hemoglobin
NT-proBNP
Albumin
BUN
eGFR
Glucose
HbA1c
hs-Troponin T
LDL
Platelets
WBC
IVSd
LVPWd
LVIDd
LVEDVi
LVESVi
Wall-thickness grade
IVSd > 15 mm
Echo EF
LV stroke volume index
Global longitudinal strain
LV systolic grade
E/A
E/e'
e' septal
e' lateral
LA volume index
Diastolic grade
LA size grade
LV diastolic dysfunction
TAPSE
RV S'
RVSP
RV diameter
RA pressure
RV size grade
RV systolic grade
AV Vmax
AV mean gradient
TR peak gradient
AS grade
AR grade
MR grade
TR grade
Mod/severe MR
Aortic root
Vitals, core labs
Other labs
LV structure
LV function
Diastolic / LA
RV / pulmonary
Valves / aorta
0.0
0.1
0.2
0.3
|SMD| (median across 32 trials)
B
PS-CVD5 ± ECG, by domain
All 54 (54)
Vitals, core labs (9)
Other labs (10)
LV structure (7)
LV function (4)
Diastolic / LA (8)
RV / pulmonary (7)
Valves / aorta (9)
0.1
0.2
Mean |SMD| (mean across 32 trials)
C
PS ladder (fewer → more covariates)
red: relative change in mean
|SMD| with + ECG vs PS alone
Mean |SMD| (32 trials*)
0.15
0.14
0.13
0.12
0.11
0.10
0.09
0.08
−11.3%
−8.8%
−5.0%
−3.2%
−7.6%
PS-Demo
PS-CVD5
PS-CVD10
PS-Clinical
(no LVEF, MI) hdPS
* hdPS + CLMBR-T + ECG: 31 trials (failed cell)
† hdPS + CLMBR-T is raised by one small trial (PROTECT AF; near-separation). Without it: hdPS 0.114, + CLMBR-T 0.113 (-0.6%, 31 trials).
D
PS-CVD5 + ECG vs PS-CVD5, per trial
20 of 32 trials improved
mean +3.6 points; P = .009
Change in % of items
with |SMD| < 0.1
20
15
10
5
0
−5
−10
Trials (sorted)